



Research Internship Abroad (BEPE/FAPESP)

**Remote Sensing Anomaly Detection in Time-Series
Imagery
and Application Development in the Agricultural
Context**

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Brazil Advisor: Dr. Rubens Augusto Camargo Lamparelli

UK Supervisor: Dr. Jefersson Alex dos Santos

UK Co-Supervisor: Prof. Po Yang

Fellow: Fernando Kenzo Imami Fugihara

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ABSTRACT

The use of plastic in agriculture has increased significantly in recent years, bringing both benefits and environmental challenges. While agricultural plastics improve crop yields and resource efficiency, they also lead to the accumulation of plastic waste in rural areas. Remote Sensing (RS) data, combined with advanced machine learning and computer vision techniques, provide an effective means to monitor plasticulture dynamics. Therefore, this research internship aims to explore RS anomaly detection (RSAD) techniques in time-series imagery and apply them to agricultural monitoring, particularly in detecting subtle cases that involve spectral changes such as material deterioration and pest-related disturbances. Therefore, I'll join researchers at the University of Sheffield to learn RSAD techniques in the agriculture context and then explore whether they can be applied in plasticulture. In parallel, I'll collaborate with researchers at the University of Sheffield and contribute to the PEZEGO pest-management app. The internship will provide hands-on experience in scalable application design, app optimization, and model integration, which can enhance our ongoing application, GeoHuman. The University of Sheffield was chosen due to its internationally recognized expertise in application development, machine learning, computer vision, and remote sensing. This experience will strengthen our project in Brazil by improving the accuracy and scalability of agricultural monitoring systems. Upon my return, I will disseminate the knowledge gained through workshops and collaborative activities with my research group at UNICAMP to foster innovation and capacity building in remote sensing applications.

1 INTRODUCTION

The use of plastic in agriculture, commonly referred to as Plasticulture (Dubois, 1978; Levin et al., 2007), has increased steadily in recent years (Perilla & Mas, 2019). According to Sintim & Flury (2017), the global annual usage of plastic films in land-based farming is estimated at 7.4 million tons, mostly from greenhouses and mulching (Food and Agriculture Organization, 2021). As plastics become increasingly prevalent in rural areas, concerns about agricultural plastic waste (APW) are escalating (Figure 1). Plastic waste has gained worldwide attention, prompting discussions among numerous countries and international organizations about potential solutions. These conversations include various aspects, including the use of plastic in agriculture (European Commission, 2019; Food and Agriculture Organization, 2021; Hofmann et al., 2023; UNEP, 2018; 2022). Therefore, mapping plasticulture sites is crucial, and one effective method involves using Remote Sensing (RS) data combined with advanced computer vision (CV) algorithms trained to classify RS images.

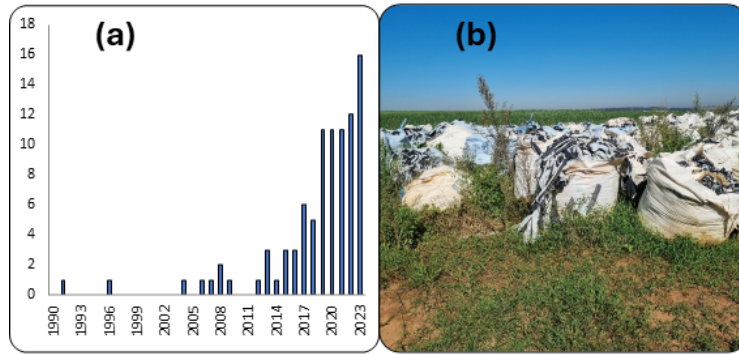


Figure 1: Growth in publications about agri-plastic remote detection. (b) APW deposited on a farm in Brazil awaiting correct disposal. Sometimes producers face difficulties finding someone to collect the plastic. Search in Web of Science on October 11th of 2023 with the query TS= ((agricultural plastic OR agro-plastic OR agroplastic OR agri-plastic OR agriplastic OR plasticulture) AND (remote sensing OR classification OR mapping OR detection) AND (greenhouse OR mulch*)).

Monitoring plasticulture requires careful attention during critical crop stages, especially those involving pest control and the post-harvest period, when the plastic cover reappears in RS imagery. These transitions can cause spectral variations detectable in RS data (Dubucq et al., 2020; Holt et al., 2025). Examples of such variations include those caused by the temporary use of pest-protection nets (Figure 2a) and the reappearance or gradual change of plastic (Figure 2b). Over time, factors such as sunlight exposure, temperature changes, and mechanical stress may also alter the material properties of agricultural plastics, subtly impacting their spectral signatures (Dubucq et al., 2020). Additionally, the use or removal of pest-control nets can also alter the scene's reflectance, introducing spectral variations detectable in RS data, verified through our field observations. In this context, Remote Sensing Anomaly Detection (RSAD) offers a valuable framework for identifying these subtle and unexpected spectral deviations from standard temporal patterns, thereby facilitating the continuous monitoring of plasticulture activities and related farming practices (Kwon & Nasrabadi, 2005).



Figure 2: Subtle dynamics of agricultural plastic use in Mossoró, RN, Brazil. a) The temporary use of a pest-control net to shield the crops. b) The reappearance and presence of the underlying plastic mulch after harvest. Photographs taken during a field visit by our research group.

RSAD has been widely studied (Racetin & Krtalić, 2021; Guo et al., 2016) and successfully applied to various domains, including environmental monitoring (Guo et al., 2016), urban mapping (Rejas Ayuga et al., 2016), and disaster detection (Anand et al., 2022). At the University of Sheffield, I will collaborate with researchers experienced in anomaly detection (AD) modeling and its applications in the agricultural sector. We will explore the potential of applying RSAD techniques to monitor the plasticulture cycle using time series data, particularly to detect subtle deviations such as plastic deterioration and pest control practices.

AD models that rely on RS imagery present significant challenges for agricultural detection due to variable illumination, heterogeneous and complex landscapes, and seasonal or phenological variations (Fu et al., 2025). A crucial step for applying RSAD techniques to time-series data is the effective preprocessing of satellite imagery, which ensures that the model outputs are reliable and accurate. One of the problems is the cloud cover, which obscures or contaminates surface reflectance and is especially evident in long-term data (Gawlikowski et al., 2022; King et al., 2013; Wen et al., 2001). In my ongoing research in Brazil, I have worked extensively on preprocessing techniques, including cloud masking, multimodal fusion, and geographic target analysis, to improve the detection of agricultural plastics. Building on this background, engaging with researchers at Sheffield will allow me to integrate preprocessing with advanced RSAD approaches.

One of the main challenges the circular economy group faced was collecting sufficient data to train CV algorithms for effective plasticulture monitoring. Given this context and considering the limited funding and manpower for field data collection, our team developed a web-based application (GeoHuman) featuring an intuitive user interface to facilitate rapid user adoption (Bödi & Kaulich, 1992). Our app, currently in development, generates high-quality RS-labeled datasets by applying CV models and preprocessing techniques in a pipeline. This work is part of the FAPESP-Braskem project (2021/052251-8). However, the application still faces significant challenges, including the need to scale efficiently to support a large number of users, reduce model prediction time, and support only local deployment. These bottlenecks currently hinder the system’s usability for broader agricultural applications.

Considering the current challenges of our research in Brazil, I intend to establish a collaboration with researchers at the University of Sheffield, who have extensive experience in developing soft-

were applications for agricultural monitoring and management. Working with them will allow me to learn advanced methodologies in scalable system design, mobile app development, and model optimization. By acquiring these techniques and incorporating them into the ongoing development of our application, I will overcome the existing limitations and significantly enhance its efficiency, scalability, and quality in Brazil.

This research internship abroad has one main goal: to increase knowledge through collaboration with researchers at the University of Sheffield, focusing on learning RSAD techniques and developing applications. We will explore whether RSAD in time series can be used in the agricultural context. These techniques could be applied in plasticulture, particularly in mulched crops. I'll also apply software development techniques to our app, GeoHuman. Therefore, this project aims to improve my skills in these areas and gain hands-on experience with a different team. The graphical abstract of the project is presented in Figure 3.

Figure 3. Graphical abstract of the BEPE/FAPESP project.

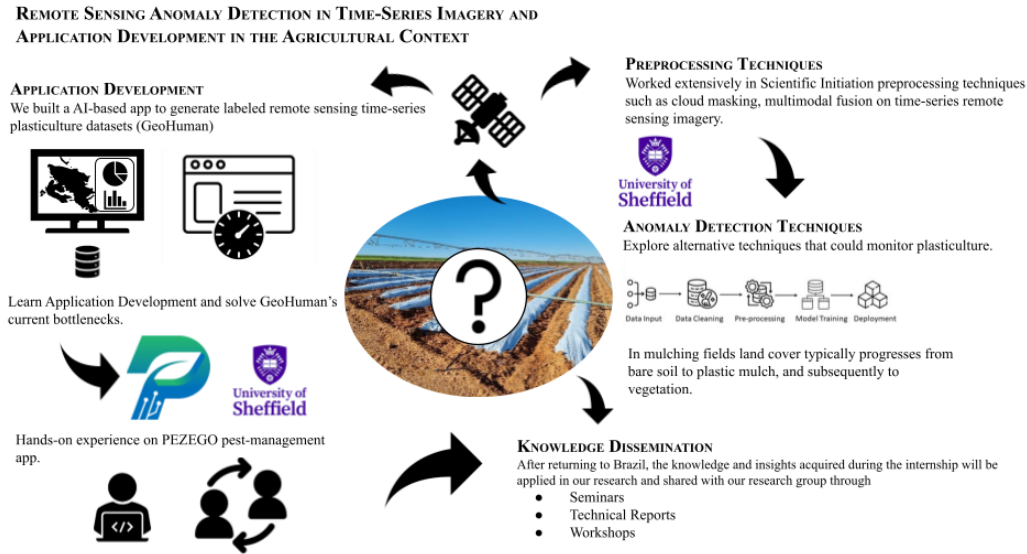


Figure 3: Graphical Abstract of BEPE.

2 METHODOLOGY

Figure 4 illustrates the conceptual framework of FAW infestation detection. We first detect the anomaly in S2 time-series (FAW infestation), then we model an anomaly detection algorithm to identify such events, and finally we can create alerts for farmers to take action.

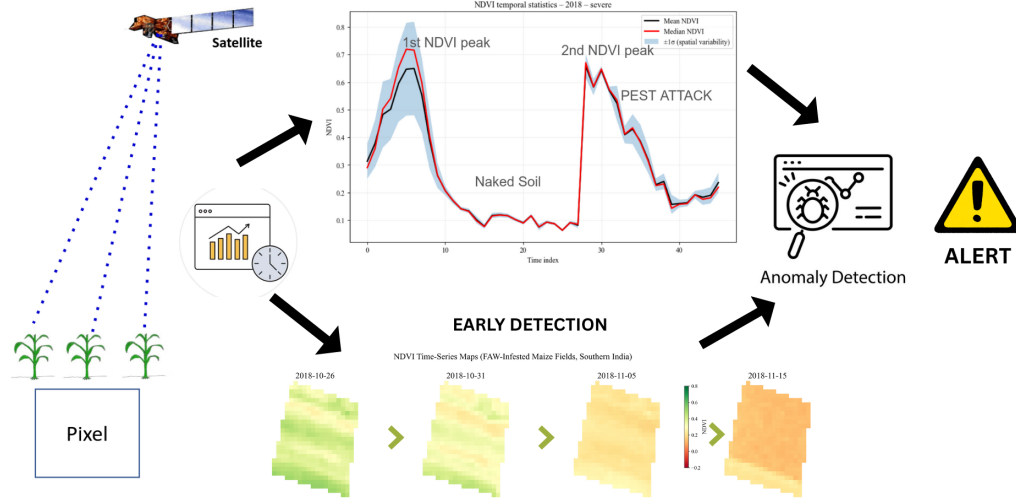


Figure 4: Conceptual framework of FAW infestation detection using Sentinel-2 time-series analysis.

2.1 DATA AND STUDY REGIONS

This research utilizes Harmonized Sentinel-2 (S2) Level-2A surface reflectance imagery, accessed through Google Earth Engine (GEE). The Copernicus Sentinel-2 mission consists of two polar-orbiting satellites (2A and 2B) operating in sun-synchronous orbit at 786 km altitude, phased 180° apart.

2.1.1 KURNOOL AND GADWAL DISTRICTS, SOUTHERN INDIA

Prabhakar et al. (2022) conducted field surveys in Kurnool and Gadwal districts of Southern India to map FAW infestations in maize crops (sorghum). In his field surveys, he mapped the regions as healthy, low, medium, and severe infestation levels. Based on that, we selected one field classified as severe infestation for our proof of concept. The Prabhakar et al. (2022) field survey confirmed that the FAW infestation occurred between October 26, 2018 and November 15, 2018. Figure 5 shows the agricultural landscape characteristics of these districts.

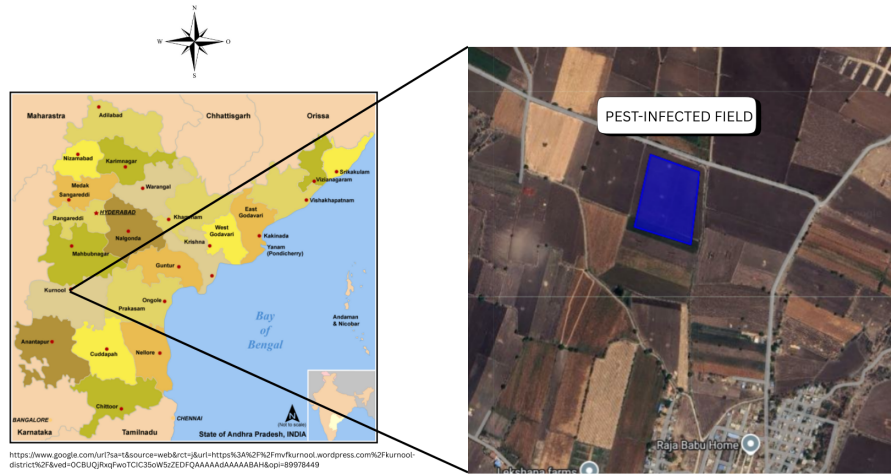


Figure 5: Kurnool and Gadwal districts, Southern India, showing maize cultivation areas.

2.1.2 EJURA, GHANA, WEST AFRICA

Figure 6 presents the diverse agricultural landscape of Ejura.

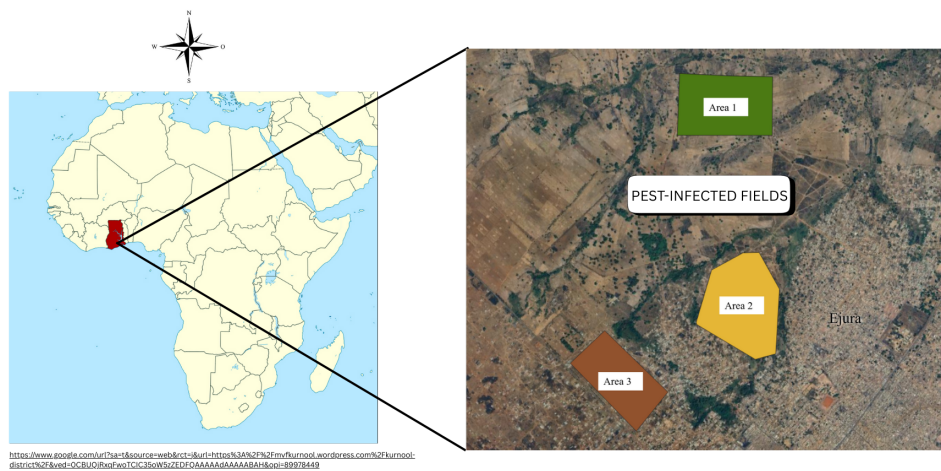


Figure 6: Ejura study region, Ghana, West Africa, showing diverse crop systems.

2.2 TIME-SERIES ANALYSIS

2.2.1 TIME-SERIES REPRESENTATION

In this study, we used a Univariate Time-Series (UTS) representation, where each field is represented by a single spectral index (e.g., NDVI) observed over time. This approach focuses on capturing the

temporal evolution and phenological patterns of a single vegetation characteristic. Given a field observed over t timestamps, the series X is defined as a sequence of scalar values:

$$X = (x_1, x_2, \dots, x_t) \quad (1)$$

Where each x_i represents the value of the spectral index at time i . Unlike multivariate approaches, this representation simplifies the analysis by focusing on the variance and trend of a single physical parameter to detect anomalies or growth stages.

2.2.2 TIME SERIES COMPONENTS

An important technique when using time series is that we can divide the data into four components: the Trend, the Seasonality, the Cyclicity and the Noise. Depending on the problem and context, these components can be analysed separately or combined to better understand the underlying patterns. In agriculture, seasonality and cyclicity are often the most informative, as crop growth and environmental conditions follow recurring patterns. In contrast, in financial markets, the trend component typically provides more insight than the others, especially for long-term analysis. The following sections provide the definitions of each component.

2.2.3 ANOMALIES IN TIME-SERIES

According to Hawkins et al (1980) an anomaly is a data point or sequence that deviates from the general data distribution. In a time-series dataset, only a small fraction contains anomalies, where the majority of the data points follow expected patterns. There are different types of anomalies in time-series including global, trend, cycle and seasonal.

In our context, the type of anomaly that we expect to find is the cycle anomaly, which is a deviation from the expected cyclic pattern. For example, in agricultural monitoring, a cycle anomaly could indicate an unexpected drop in vegetation health during a period when crops are typically thriving, potentially signaling pest infestations or disease outbreaks.

2.2.4 TIME SERIES SMOOTHING

Time series smoothing is a technique used to reduce noise and improve data quality, making underlying patterns easier to analyse. In this study, we applied Triple Exponential Smoothing (TES) / Holt–Winters, which effectively models the level, trend, and seasonality components commonly present in agricultural satellite time series. A.1 presents different smoothing techniques and its mathematical formulations.

2.3 PREPROCESSING SENTINEL-2 IMAGERY

In remote-sensing data, there is a lot of different informations that deviate from normality, such as sensor noise, atmospheric effects, cloud cover, seasonal variations, and landscape complexity (Fu et al., 2025). The most meaningful deviations are usually those that are significantly different from the norm.

Preprocessing is a crucial step in machine learning and data pipelines (Tawakuli et al., 2025). It employs various techniques to ensure that the data fed into a model is clean, consistent, and informative, which directly influences its performance and reliability (Zheng et al., 2021). This is especially important in RSAD modeling. In RS imagery, data quality can be compromised by factors such as Earth’s geometric effects resulting from its curvature and rotation, atmospheric conditions like cloud cover, sensor noise, seasonal variations, and landscape complexity (Fu et al., 2025).

2.3.1 SPECTRAL INDICES

Studies from Dzurune et al. (2025), Bilintoh et al. (2019), Prabhakar et al. (2022), have demonstrated the effectiveness of Normalized Difference Vegetation Index (NDVI) Enhanced Vegetation Index (EVI) and Leaf Area Index (LAI), respectively, in detecting pest infestations. Table 1 presents the spectral indices used in this research, along with their mathematical formulations.

Table 1: Spectral vegetation indices and corresponding equations using Sentinel-2 bands.

Index	Equation	Reference
NDVI	$\frac{NIR - Red}{NIR + Red}$	Rouse et al., 1974
EVI	$2.5 \frac{NIR - R}{NIR + 6R - 7.5B + 1}$	Huete et al., 1997
LAI	$3.618 \times EVI - 0.118$	Boegh et al., 2002

2.4 ANOMALY DETECTION MODEL

In order to detect pest attacks, we calculated the ration between the NDVI and time periods, if a certain decline happened outside of the harvest period or other known events, then we can consider this as an anomaly and potentially related to pest attacks. We also can use the LAI time-series maps to spatially identify areas of the field that are more affected by the pest infestation, which can help farmers to take action in specific areas of the field.

3 RESULTS AND DISCUSSION

3.1 FAW INFESTATION DETECTION IN SOUTHERN INDIA WITH SENTINEL-2

To validate wheather S2 time-series can capture FAW infestation, we followed the methodology proposed by Prabhakar et al. (2022) which analyzed S2 time-series detect FAW infestation in maize crops in Southern India. Prabhakar et al. (2022) utilized ground truth data from field surveys to clas-sify infestation severity levels (healthy, mild, moderate, severe) and correlated these with spectral signatures and vegetation index time-series derived from S2 imagery. In its field surveys, Prabhakar et al.(2022) mapped FAW infestation dated from October 26, 2018 to November 15, 2018, these dates and a severe field were selected for our proof of concept.

3.1.1 SPECTRAL SIGNATURES OF PEST INFESTATION

Based on the severity classification framework proposed by Prabhakar et al. (2022),FAW damage is categorized into four levels: healthy (Grade-1), low (Grade-2), medium (Grade-3), and severe (Grade-4) infestation. In our study, we used the severe cases of the Prabhakar et al. (2022) field survey data available only at two key moments:a pre-infestation survey, representing healthy crop conditions, and a post-infestation survey,corresponding to severe FAW damage.These two surveys were used as temporal anchors to characterize the spectral responseof maize crops before and after the pest outbreak. The resulting spectral signatures associated with each severity level are presented in Figure7.

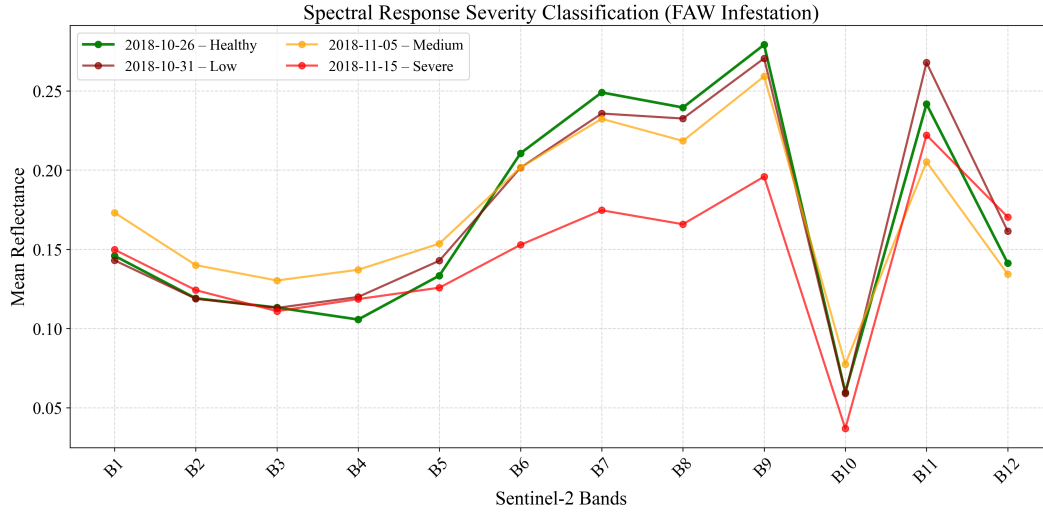


Figure 7: Spectral signatures of maize crops under different severity levels of Fall Armyworm infestation in Southern India.

Based on the mean spectral response of the field illustrated in Figure 7, we can interpret its results based on each spectral band characteristics. Near-Infrared (B8): carries information about leaf structure. Damage from pest feeding reduces reflectance in this band. Blue (B2) and Red (B4) carry chlorophyll information. Chlorophyll absorbs these wavelengths; reduction in absorption indicates decreased photosynthetic activity, often caused by pest feeding on leaves. Shortwave Infrared (B10-B12) are sensitive to water content. Pest-damaged plants often exhibit water stress, detectable in these bands.

Harvest events cause abrupt transitions from high NDVI (green, healthy) to low NDVI (senescence, yellow/brown), which must be distinguished from pest-related declines. Pest-related NDVI declines are more gradual and associated with specific temporal patterns linked to pest life cycles and feeding behavior.

3.1.2 TIME-SERIES ANALYSIS

Figure 8 presents the NDVI time-series of the FAW-infested field in Southern India for the year of 2019, while Figure 9 shows the NDVI time-series of the same field for the year of 2018. The temporal patterns in these time-series reveal significant differences between the two years, particularly during the period of FAW infestation. In 2018, we observe a clear decline in NDVI values corresponding to the infestation period, indicating a reduction in vegetation health and biomass due to pest damage. In contrast, the 2019 time-series shows a more stable NDVI pattern, suggesting healthier vegetation conditions without significant pest-related stress. These observations highlight the potential of Sentinel-2 time-series analysis for monitoring crop health and detecting pest infestations in agricultural fields.

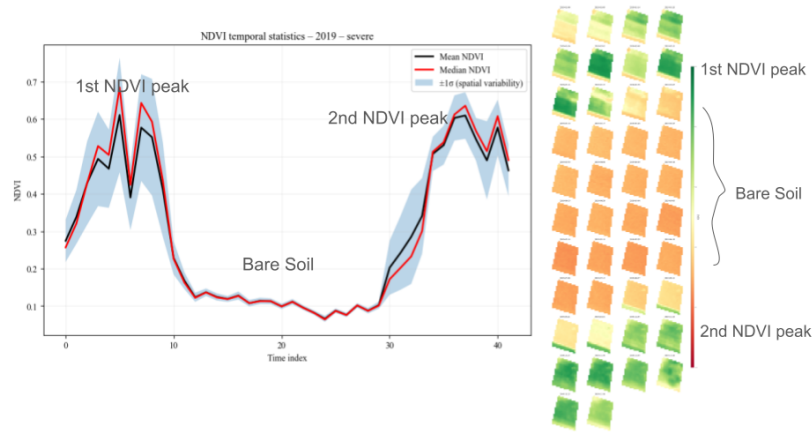


Figure 8: NDVI time series of 2019 in Southern India.

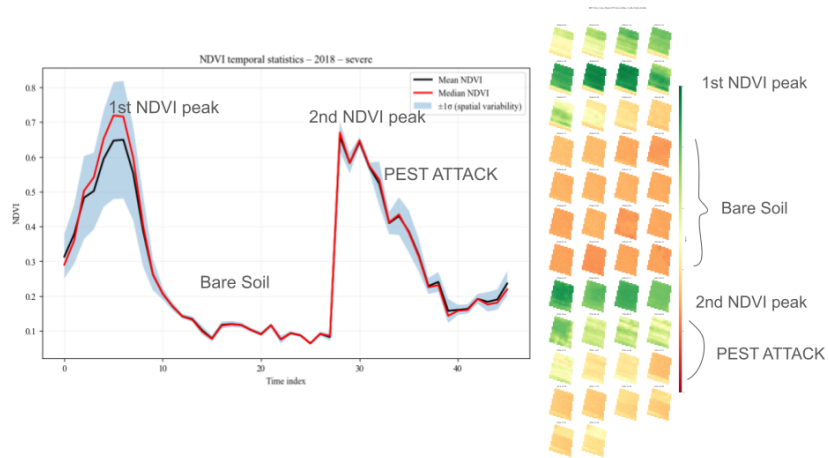


Figure 9: NDVI time series of 2018 in Southern India.

Figure 10 show the LAI time-series maps of the FAW-infested field in Southern India. We observe a clear temporal decline in LAI values across the field, corresponding to the period of FAW infestation. The spatial patterns in these maps highlight the extent and severity of the damage caused by the pest, with lower index values indicating areas of more severe infestation and biomass loss.

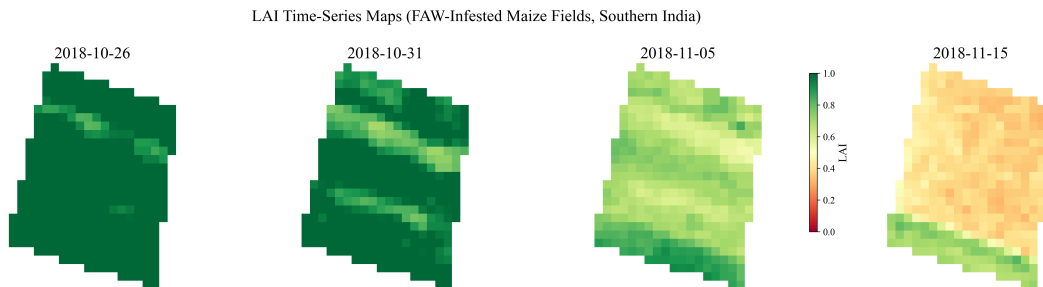


Figure 10: LAI time series of FAW-infested in Southern India.

Based on the results presented in Figures 8, 9, and 10, we can conclude that Sentinel-2 time-series analysis is a valuable tool for monitoring crop health and detecting pest infestations. The observed decline in NDVI and LAI values during the FAW infestation period indicates significant stress on the vegetation, which can be effectively captured through remote sensing techniques. These findings underscore the potential of using satellite imagery for early detection of pest outbreaks, enabling timely interventions to mitigate crop damage and improve agricultural productivity.

3.2 FAW INFESTATION DETECTION IN GHANA WITH SENTINEL-2

3.3 ANOMALY DETECTION & EARLY ALERTS

To detect anomalies in the Sentinel-2 time-series, we applied a simple threshold-based approach to identify significant declines in NDVI values that deviate from expected seasonal patterns. By analyzing historical NDVI data for the field, we established a baseline of normal vegetation dynamics and set thresholds to flag potential anomalies indicative of pest infestations. The results showed that during the FAW infestation period, there were significant drops in NDVI values that exceeded the established thresholds, confirming the presence of anomalies associated with pest damage. This approach demonstrates the feasibility of using Sentinel-2 time-series analysis for early detection of pest outbreaks, allowing farmers to take proactive measures to protect their crops.

4 APPLICATION DEVELOPMENT

In parallel to the RSAD modeling, I participated in app development testing and remote sensing model integration for the PEZEGO pest-management app. This experience provided insights into scalable application design, optimization techniques, and model deployment strategies, which can be applied to enhance our ongoing project, GeoHuman. The PEZEGO app focuses on providing actionable insights to farmers for pest management, and my involvement in this project allowed me to contribute to the development of user-friendly interfaces and efficient data processing pipelines.

5 FUTURE WORK

Future work will prioritize a dedicated field survey campaign to validate the Sentinel-2 anomaly detection results and strengthen the reliability of the conclusions. Ground observations will be collected for both plastic mulch deterioration and FAW infestation severity, including georeferenced samples, visual scoring, and photographic records at multiple dates during the crop cycle.

These in situ measurements will be used to (i) calibrate anomaly thresholds, (ii) compare satellite-derived indicators (NDVI, LAI, PI) with observed field conditions, and (iii) quantify model performance through confusion matrices and class-wise metrics (precision, recall, F1-score). This validation step is expected to provide more concrete evidence of the method's practical applicability and improve the robustness of early-warning recommendations for farmers.

6 CONCLUSION

7 INTERNSHIP EXPERIENCE

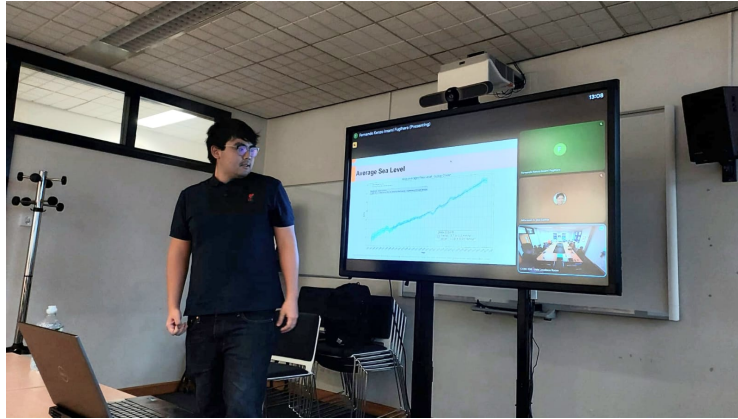


Figure 11: Presenting my project results in a weekly seminar for the Computer Vision Group at the University of Sheffield.



Figure 12: Presenting my project results in a weekly seminar for the Computer Vision Group at the University of Sheffield.

REFERENCES

- P. Anand, M. Sharma, and A. Saroliya. Anomaly detection in disaster recovery: A review, current trends and new perspectives. In *Proceedings of 5th International Conference on Contemporary Computing and Informatics (IC3I)*, 2022. doi: 10.1109/IC3I56241.2022.10072941.
- R. A. Bödi and T. W. Kaulich. Intuitive user interfaces (iui): a case starting point for design and programming. *Computer Methods and Programs in Biomedicine*, 37(2), 1992. doi: 10.1016/0169-2607(92)90087-N.
- P. Dubois. *Plastics in agriculture*. Applied Science Publishers, 1978.
- D. Dubucq, R. Jatiault, P. Boymond, and A. Rusquet. Remote sensing detection of plastic waste: recent improvements and remaining challenges, 2020.
- European Commission. A circular economy for plastics – insights from research and innovation to inform policy and funding decisions. Technical report, European Commission, 2019.
- Food and Agriculture Organization. Assessment of agricultural plastics and their sustainability: A call for action. Technical report, FAO, 2021.
- Y. Fu, H. Peng, T. Zhao, Y. Li, J. Peng, and D. Zhang. Lightweight remote sensing change detection with progressive multi scale difference aggregation. *Scientific Reports*, 15(1):30203, 2025. doi: 10.1038/s41598-025-10972-5.
- J. Gawlikowski, P. Ebel, M. Schmitt, and X. X. Zhu. Explaining the effects of clouds on remote sensing scene classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15, 2022. doi: 10.1109/JSTARS.2022.3221788.
- Q. Guo, R. Pu, L. Gao, and B. Zhang. A novel anomaly detection method incorporating target information derived from hyperspectral imagery. *Remote Sensing Letters*, 7(1), 2016. doi: 10.1080/2150704X.2015.1101177.
- T. Hofmann, S. Ghoshal, N. Tufenkji, J. F. Adamowski, S. Bayen, Q. Chen, P. Demokritou, M. Flury, T. Hüffer, N. P. Ivleva, R. Ji, R. L. Leask, M. Maric, D. M. Mitrano, M. Sander, S. Pahl, M. C. Rillig, T. R. Walker, J. C. White, and K. J. Wilkinson. Plastics can be used more sustainably in agriculture. *Communications Earth and Environment*, 4(1), 2023. doi: 10.1038/s43247-023-00982-4.
- Z. K. Holt, S. D. Khan, and D. F. Rodrigues. Hyperspectral remote sensing as an environmental plastic pollution detection approach to determine occurrence of microplastics in diverse environments. *Environmental Pollution*, 377:126426, 2025. doi: 10.1016/j.envpol.2025.126426.
- M. D. King, S. Platnick, W. P. Menzel, S. A. Ackerman, and P. A. Hubanks. Spatial and temporal distribution of clouds observed by modis onboard the terra and aqua satellites. *IEEE Transactions on Geoscience and Remote Sensing*, 51(7), 2013. doi: 10.1109/TGRS.2012.2227333.
- H. Kwon and N. M. Nasrabadi. Kernel rx-algorithm: A nonlinear anomaly detector for hyperspectral imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 43(2), 2005. doi: 10.1109/TGRS.2004.841487.
- N. Levin, R. Lugassi, U. Ramon, O. Braun, and E. Ben-Dor. Remote sensing as a tool for monitoring plasticulture in agricultural landscapes. *International Journal of Remote Sensing*, 28(1), 2007. doi: 10.1080/01431160600658156.
- G. A. Perilla and J. F. Mas. High-resolution mapping of protected agriculture in mexico, through remote sensing data cloud geoprocessing. *European Journal of Remote Sensing*, 52(1), 2019. doi: 10.1080/22797254.2019.1686430.
- I. Racetin and A. Krtalić. Systematic review of anomaly detection in hyperspectral remote sensing applications. *Applied Sciences*, 11(11), 2021. doi: 10.3390/app11114878.

- J. G. Rejas Ayuga, R. Martínez Marín, M. Marchamalo Sacristán, J. Bonatti, and J. C. Ojeda. Hyperspectral anomaly detection in urban scenarios. In *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, volume XLI-B7, 2016. doi: 10.5194/isprs-archives-xli-b7-111-2016.
- H. Y. Sintim and M. Flury. Is biodegradable plastic mulch the solution to agriculture's plastic problem? *Environmental Science and Technology*, 51(3), 2017. doi: 10.1021/acs.est.6b06042.
- UNEP. Mapping of global plastics value chain and plastics losses to the environment. Technical report, United Nations Environment Programme, 2018.
- UNEP. Resolution 5/14. end plastic pollution: Towards an international legally binding instrument. Technical report, United Nations Environment Programme, March 2022.
- G. Wen, R. F. Cahalan, S. C. Tsay, and L. Oreopoulos. Impact of cumulus cloud spacing on landsat atmospheric correction and aerosol retrieval. *Journal of Geophysical Research Atmospheres*, 106 (D11), 2001. doi: 10.1029/2001JD900159.

A APPENDIX

A.1 TIME-SERIES SMOOTHING TECHNIQUES

Smoothing techniques are essential preprocessing steps that remove noise from time-series data while preserving important patterns. This is particularly critical when comparing pixel time-series or applying distance-based algorithms.

Consider two pixels with similar underlying patterns but different noise characteristics. Without smoothing, distance-based clustering algorithms may incorrectly classify these pixels as dissimilar due to noise rather than genuine differences in behavior. Smoothing addresses this issue by enhancing signal-to-noise ratio.

The choice of smoothing method depends on the structural characteristics of the time-series, as summarized in Table A.1.

Table A.1: Smoothing algorithm applicability for different time-series structures

Algorithm	Level	Trend	Seasonality	Parameters
Single HWES	Yes	No	No	α
Double HWES	Yes	Yes	No	α, β
Triple HWES	Yes	Yes	Yes	α, β, γ

Table A.2 defines the variables used in smoothing formulations.

Table A.2: Variables utilized in smoothing models

Symbol	Description
X	Observation
S	Smoothed observation
B	Trend factor
C	Seasonal index
F	Forecast at m periods ahead
α	Data smoothing factor, $\alpha \in (0, 1)$
β	Trend smoothing factor, $\beta \in (0, 1)$
γ	Seasonal change smoothing factor, $\gamma \in (0, 1)$
ϕ	Damped smoothing factor, $\phi \in (0, 1)$
t	Time period index

A.1.1 MOVING AVERAGE AND WEIGHTED AVERAGE

These methods compute the future value as the average (or weighted average) of k previous values. While useful for trend observation and feature engineering, these approaches are unsuitable for satellite time-series with trend and seasonality components.

A.1.2 SINGLE EXPONENTIAL SMOOTHING (SES)

SES models the level component only and is appropriate for stationary series without trend or seasonality. It weights recent observations more heavily based on the assumption that the future is more closely related to the recent past.

$$S_0 = X_0$$

$$S_t = \alpha X_t + (1 - \alpha)S_{t-1}, \quad t > 0, 0 < \alpha < 1$$

Application: Limited applicability to agricultural time-series, which typically exhibit both trend and seasonality.

A.1.3 DOUBLE EXPONENTIAL SMOOTHING (DES)

DES extends SES by incorporating trend, making it suitable for series with trend but without seasonality.

$$\begin{aligned} S_0 &= X_0 \\ B_0 &= X_1 - X_0 \\ S_t &= \alpha X_t + (1 - \alpha)(S_{t-1} + B_{t-1}) \\ B_t &= \beta(S_t - S_{t-1}) + (1 - \beta)B_{t-1}, \quad \alpha, \beta \in (0, 1) \end{aligned}$$

Application: Applicable to agricultural time-series when seasonality is not a dominant factor.

A.1.4 TRIPLE EXPONENTIAL SMOOTHING (TES) / HOLT-WINTERS

TES represents the most comprehensive smoothing approach, modeling level, trend, and seasonality simultaneously. This method is most appropriate for agricultural satellite time-series, which exhibit all three components.

$$\begin{aligned} S_0, F_0 &= X_0 \\ B_0 &= \frac{\sum_{i=0}^{L-1} (X_{L+i} - X_i)}{L^2} \\ S_t &= \alpha (X_t - C_{t \bmod L}) + (1 - \alpha) (S_{t-1} + \phi B_{t-1}) \\ B_t &= \beta (S_t - S_{t-1}) + (1 - \beta) \phi B_{t-1} \\ C_{t \bmod L} &= \gamma (X_t - S_t) + (1 - \gamma) C_{t \bmod L} \\ F_{t+m} &= S_t + B_t \sum_{i=1}^m \phi^i + C_{t \bmod L}, \quad \alpha, \beta, \gamma \in (0, 1) \end{aligned}$$

Application: Recommended for Sentinel-2 vegetation index time-series, which exhibit clear seasonal patterns related to crop phenology.